AI ACADEMY · LEVEL 1 · GRADE 6 · WEEK 13 · WORKBOOK

Learning from Examples

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Helpful Classroom Sorter

THE CONNECTED WEEK

One connected fitting-and-version record.

Trace candidates, explain a tie, add fitting evidence and inspect contradictory labels.

Keep one habit: Keep the fitting data, selection policy and model version together.

MAIN LESSON · SECTION 1

Recall: reconnect earlier ideas

> Retrieve before opening notes.

Retrieve and explain

- Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

- Without opening the old lesson, explain one key idea from Debug the Cause. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from If/Then Decisions. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Categories and Labels. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

My remembered idea, example and correction

My remembered idea, example and correction

▼ **Earlier wording, alternatives and picture transcripts**

Debug the Cause

If/Then Decisions

Categories and Labels

W1 · Trace each candidate separately

> List the mismatches before trusting each error total.

Your evidence record

- Heights 2, 4, 6 and 8 cm have labels short, short, tall and tall. Predict all four labels under threshold 3, then under threshold 7, using tall when height is at least the threshold.

  List the disagreeing examples and separate totals.

> **My reasoning and evidence**

MAIN LESSON · SECTION 13

W2 · Explain a tied fit

> Give a missing input where the tied candidates disagree.

Your evidence record

- Thresholds 5 and 6 both have zero disagreements on the W1 **fitting set**. The **tie rule** chooses the smaller threshold. Which is selected? Give a whole-number height where the two candidates disagree and both **prediction**s.

  Explain what the **tie rule** does not prove.

My reasoning and evidence

MAIN LESSON · SECTION 14

W3 · Refit after adding a labelled example

> Keep all earlier records and update every candidate count.

Your evidence record

- Add height 5 cm labelled short to the four **fitting** examples. Recompute the disagreement totals for thresholds 3, 5, 6 and 7. Select the candidate with the fewest disagreements, keeping the smaller-threshold **tie rule**.

  Explain why the new label is part of **fitting**.

My reasoning and evidence

MAIN LESSON · SECTION 16

W5 · Inspect identical inputs with conflicting labels

> Explain the limitation of a deterministic feature-only rule.

Your evidence record

- Two **fitting** records both have height 6 cm. One is labelled short and the other tall. A learner says a clever height-only rule can agree with both on every run.

  Explain the problem for a deterministic rule and name two checks to make before changing either record.

My reasoning and evidence

MAIN LESSON · SECTION 4

W4 · Fit a new distance task

> Keep one first attempt before hints or feedback.

Read and explain independently

Imaginary map markers have distances 1, 3, 5 and 7 units, with supplied labels near, far, far and far. Compare candidates that predict far when distance is at least 2, 4 or 6 units.

List every **prediction** and error count, select the lowest-error candidate, then predict at distance 2 before any comparison label is supplied. Limit your conclusion.

My independent explanation, trace and limits

Keep a fair record

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

▼ **Earlier wording, alternatives and picture transcripts**

Independent case: Fit a new distance task

MAIN LESSON · SECTION 9

Project · Keep the decision and its evidence

> Add this week's change to your existing project record.

Your evidence record

Create the first small labeled example set for the sorter and explain which visual differences it covers.

The project change and **evidence** I will keep

Write or draw your reasoning and evidence.

MAIN LESSON · SECTION 17

Feedback · Repair and return

> Keep the first attempt, the repair and a changed-case return.

Your evidence record

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

MAIN LESSON · SECTION 10

Finish · Judge the evidence in your work

> Point to work that supports your confidence level.

Your evidence record

- I can explain the mechanism.
- I can show evidence.
- I can handle a changed case.
- I can name a limit.

Mark each as not yet, with help or independently, and attach supporting work.

Teacher review _____ Next step _____

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Supported W2 record**

REFERENCE · SECTION 2

Reference · Supported W2 record

> This is the same W2 attempt. Keep one response record.

Use this when helpful

Use W2: Explain a tied fit for **Fitting** a **pattern** from examples. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

The step I completed without help, and the hint I needed

▼ Extra practice · Repair the explanation

REFERENCE · SECTION 3

Extra practice · Repair the explanation

> Explain the deciding fact before discussing the answer.

Use this when helpful

Use the worked case for **Fitting** a **pattern** from examples. Proposed explanation: “The lowest error count is added across candidates”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

> **My explanation before partner discussion**

▼ Reference · Original learning target

REFERENCE · SECTION 5

Reference · Original learning target

> Use this target to inspect an explanation.

Use this when helpful

Training cards show lengths 2,3 labelled S and 7,8 labelled L. Two proposed rules are 'L at 5 or above' and 'L at 7 or above'. Both fit **training**.

Choose one new labelled card length that would distinguish them and explain what remains unknown.

▼ **Extra practice · A complete case**

REFERENCE · SECTION 6

Extra practice · A complete case

> Try the question before checking the worked explanation.

Use this when helpful

Training examples label smooth leaves S and toothed leaves T. A new leaf is toothed.

Predict and state the **evidence** limit.

Show your trace, calculation or **evidence**

> **Write or draw your reasoning and evidence.**

▼ Extra practice · A changed case

REFERENCE · SECTION 7

Extra practice · A changed case

> Keep this practice separate from the reserved independent attempt.

Use this when helpful

A learner sees three red apples and no green apples, then calls a green apple not an apple.
What missing **evidence** could help? How would you check improvement?

My practice answer and reason

> **Write or draw your reasoning and evidence.**

Week 13 Test and reflect

▼ Extra practice · Design a revealing test

REFERENCE · SECTION 8

Extra practice · Design a revealing test

> Predict before tracing or testing.

Use this when helpful

Create a changed **input**, boundary case or missing-information case. Write the expected answer or safe response before testing.

Test record	My evidence
Changed input or condition	
Expected answer or fallback	
Observed result	
Why the result occurred	

▼ **Reference · Original target response space**

REFERENCE · SECTION 11

Reference · Original target response space

> This teaching case is practice rather than a fresh assessment.

Use this when helpful

Training cards show lengths 2,3 labelled S and 7,8 labelled L. Two proposed rules are 'L at 5 or above' and 'L at 7 or above'. Both fit **training**.

Choose one new labelled card length that would distinguish them and explain what remains unknown.

My explanation after practice

Record support used: none, hint or teacher explanation.

▼ **Reference · Independent-case response copy**

REFERENCE · SECTION 15

Reference · Independent-case response copy

> This duplicates W4. Reuse the same first-attempt record.

Use this when helpful

Imaginary map markers have distances 1, 3, 5 and 7 units, with supplied labels near, far, far and far. Compare candidates that predict far when distance is at least 2, 4 or 6 units.

List every **prediction** and error count, select the lowest-error candidate, then predict at distance 2 before any comparison label is supplied. Limit your conclusion.

My reasoning and evidence

▼ Extra practice · Use the worked picture

REFERENCE · SECTION 18

Extra practice · Use the worked picture

> Keep the explanation and the support you used.

Use this when helpful

A fictional classifier learns whether measured sticks are short or long.

Training lengths 2,4,8,10 cm have labels short,short,long,long.

Candidate boundaries: long at least 5 or at least 7. Lowest errors wins; tie chooses smaller boundary.

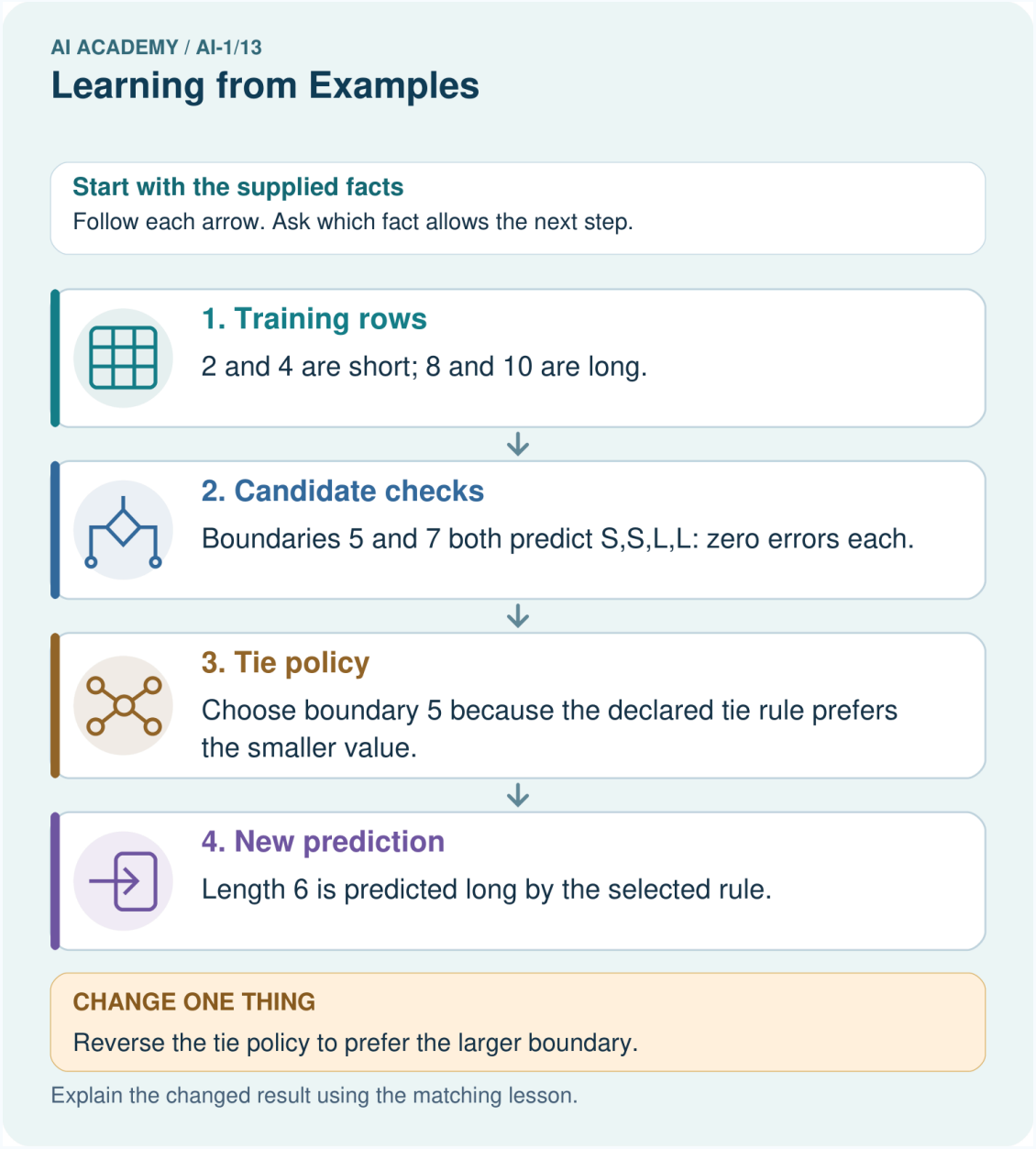
AI ACADEMY / AI-1/13

Same training errors, different new prediction

	Boundary	Training errors	Prediction at 6
Candidate 5	5	0	long
Candidate 7	7	0	short

Read a cell using both its row and column labels.

Follow the labelled picture with the matching explanation. Open the vector diagram



Follow the labelled picture with the matching explanation. Open the vector diagram

Which new measured length would reveal disagreement between the candidates?

My picture-based explanation and changed-case prediction

Support used: alone / hint / worked example. This record is practice **evidence**.

Project connection: Save candidate definitions, **training** errors and tie policy with any sorter model version.

▼ **Reference · Project portfolio record**

REFERENCE · SECTION 19

Reference · Project portfolio record

> Extend your main project response when helpful.

Use this when helpful

Project: Helpful Classroom Sorter

Current milestone: Test changed cases. Keep normal, boundary and failure cases; compare expected and observed results.

Save candidate definitions, **training** errors and tie policy with any sorter model version.

Evidence to keep	My record
This week’s input and deciding step	
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	
One error or limit and the next test	

Explain what this week adds to your project. Reuse last week's record rather than starting again.

▼ **Extra practice: pictures, repair and feedback**

REFERENCE · SECTION 20

Extra practice: pictures, repair and feedback

> Attempt each question before opening its explanation. Use one browser or paper copy.

First supported question

- A fictional classifier learns whether measured sticks are short or long.
- Which labelled step matches this detail? Length 6 is predicted long by the selected rule.
- 1. **Training** rows
- 2. New **prediction**
- 3. Tie policy
- 4. Candidate checks
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check the first explanation**

After your attempt, compare with the supported explanation: New **prediction** Length 6 is predicted long by the selected rule. There is no **training** stick between 4 and 8 to distinguish these candidates.

The tie policy chooses one consistently; it is not additional **evidence** that its **prediction** at 6 is true. Label-free **prediction** applies the selected boundary, while evaluation would require an appropriate new reference label.

Changed-case question

- Changed-case practice: Reverse the tie policy to prefer the larger boundary. Which conclusion is supported by the supplied facts?
- 1. This one example guarantees correct results for every future **input**.
- 2. We can decide the result without examining the changed **input** or the procedure.
- 3. Boundary 7 is selected and predicts short at 6.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check the changed case**

After your attempt, compare with the supported explanation: Boundary 7 is selected and predicts short at 6. The **training** fit is unchanged; only the explicit choice between tied candidates changes.

Repair and save your work

Review Debug the Cause (ai-1/12). Goal: Compare expected and observed traces to locate the first difference. Short practice: Propose **input** 0 and give both predicted **output**s without inventing an observed result..

Return to this week and explain what you changed.

Review If/Then Decisions (ai-1/10). Goal: Evaluate a condition using the current state and trace only the selected branch. Short practice: Which **input** detects the difference between the two written conditions?

Explain its entire run.. Return to this week and explain what you changed.

Review Categories and Labels (ai-1/7). Goal: Apply a category definition using explicit **evidence** and boundary rules. Short practice: Explain why B originally was revise rather than review, then label its changed version..

Return to this week and explain what you changed.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test

▼ **Earlier wording, alternatives and picture transcripts**

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Keep your completed work

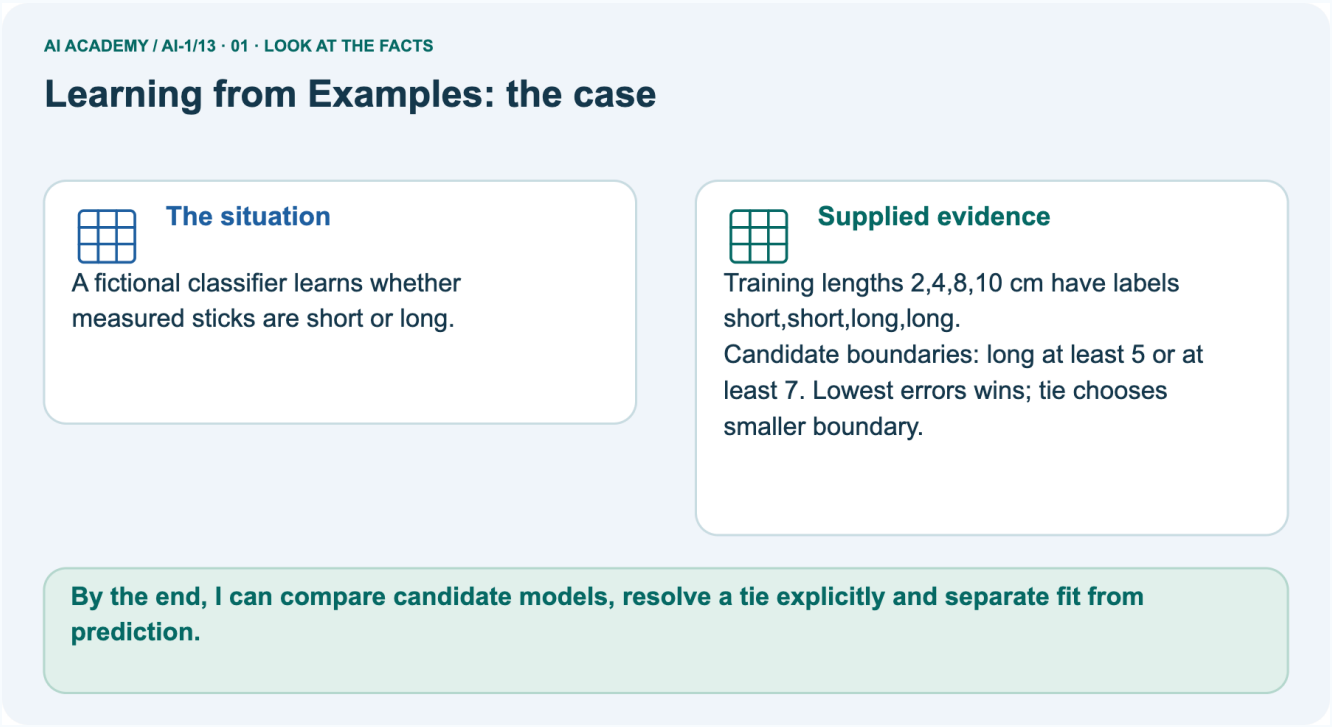
▼ **Reference · Six-picture response sheets**

REFERENCE · SECTION 21

Reference · Six-picture response sheets

> Choose the image that helps rather than repeating all six.

01. Learning from Examples: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

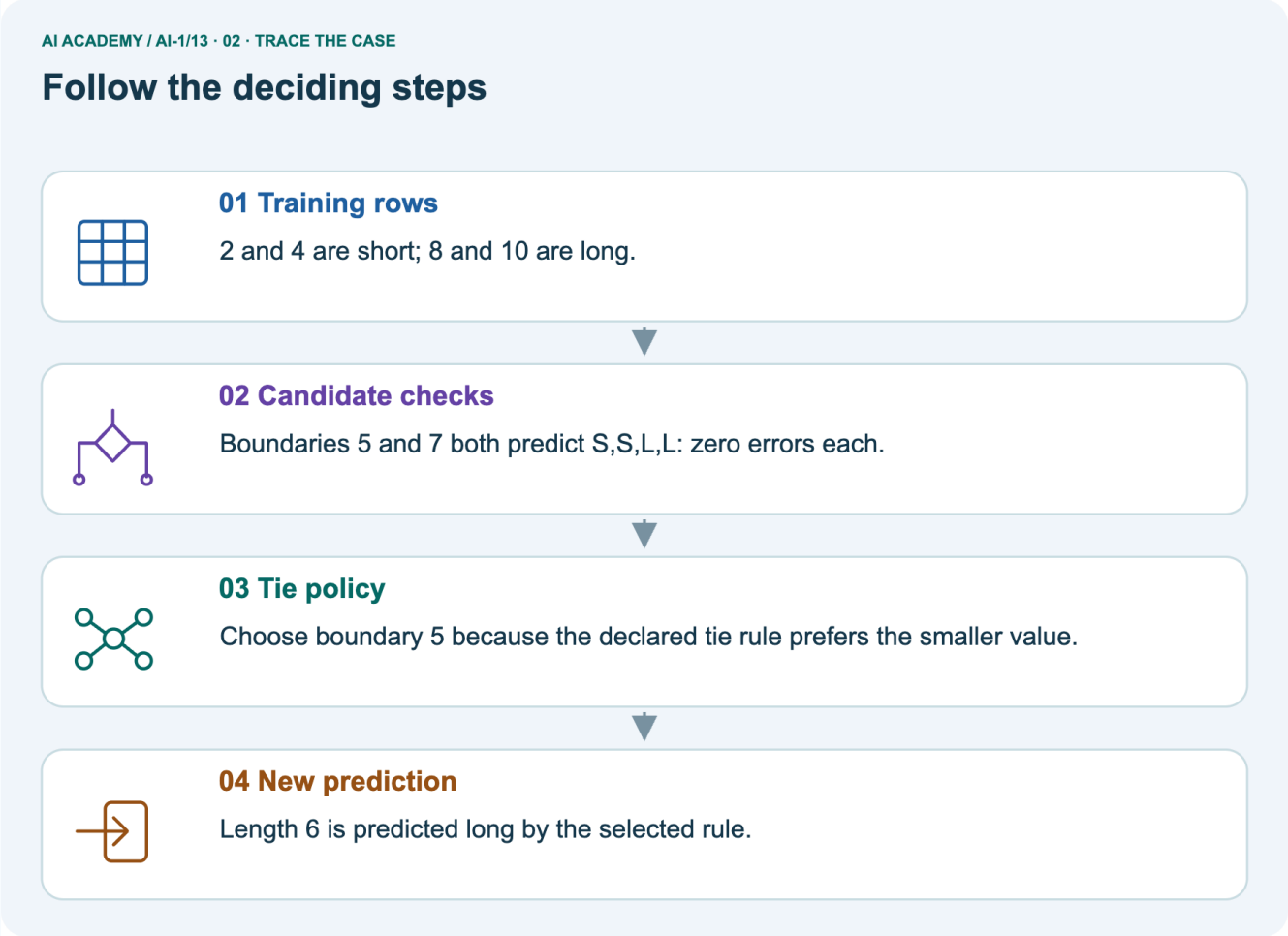
A fictional classifier learns whether measured sticks are short or long. Supplied facts: **Training** lengths 2,4,8,10 cm have labels short,short,long,long. Candidate boundaries: long at least 5 or at least 7.

Lowest errors wins; tie chooses smaller boundary.

Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



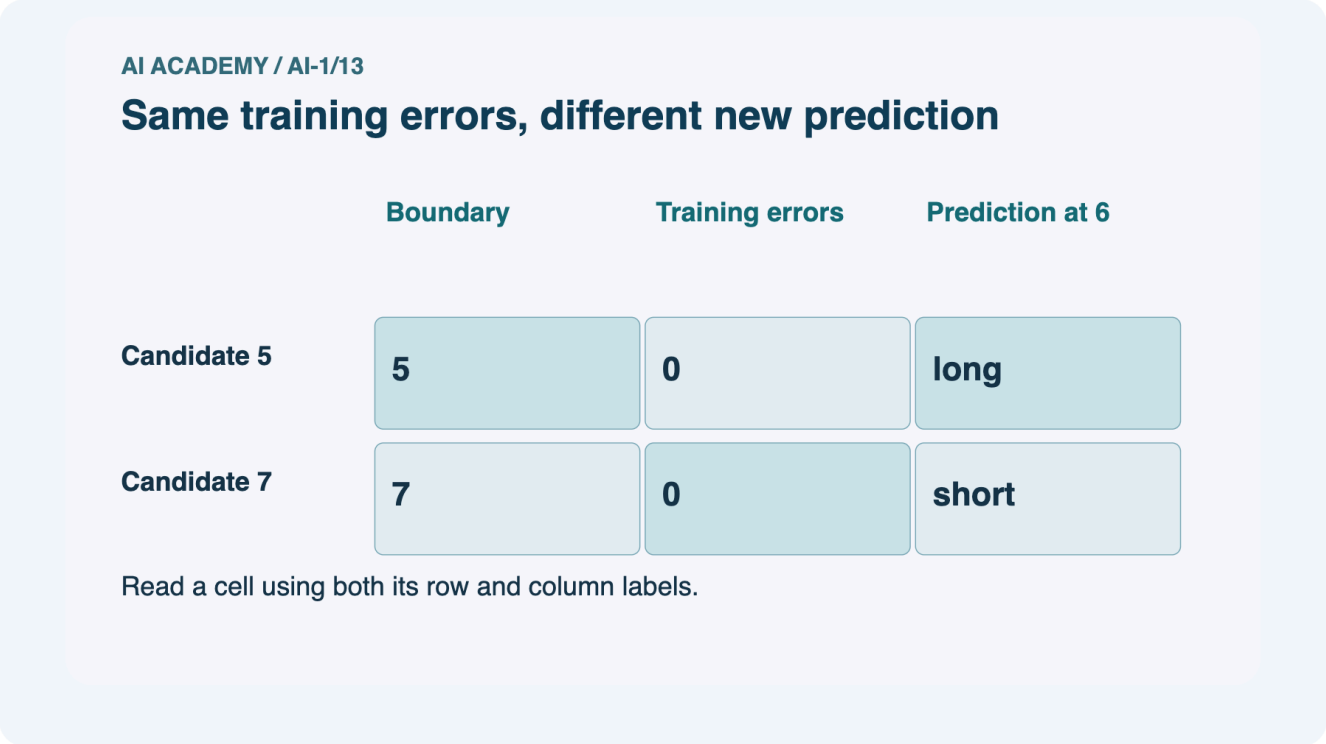
Follow the labelled picture with the matching explanation. Open the vector diagram

Training rows: 2 and 4 are short; 8 and 10 are long. → Candidate checks: Boundaries 5 and 7 both predict S,S,L,L: zero errors each. → Tie policy: Choose boundary 5 because the declared **tie rule** prefers the smaller value. → New **prediction**: Length 6 is predicted long by the selected rule.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Same training errors, different new prediction



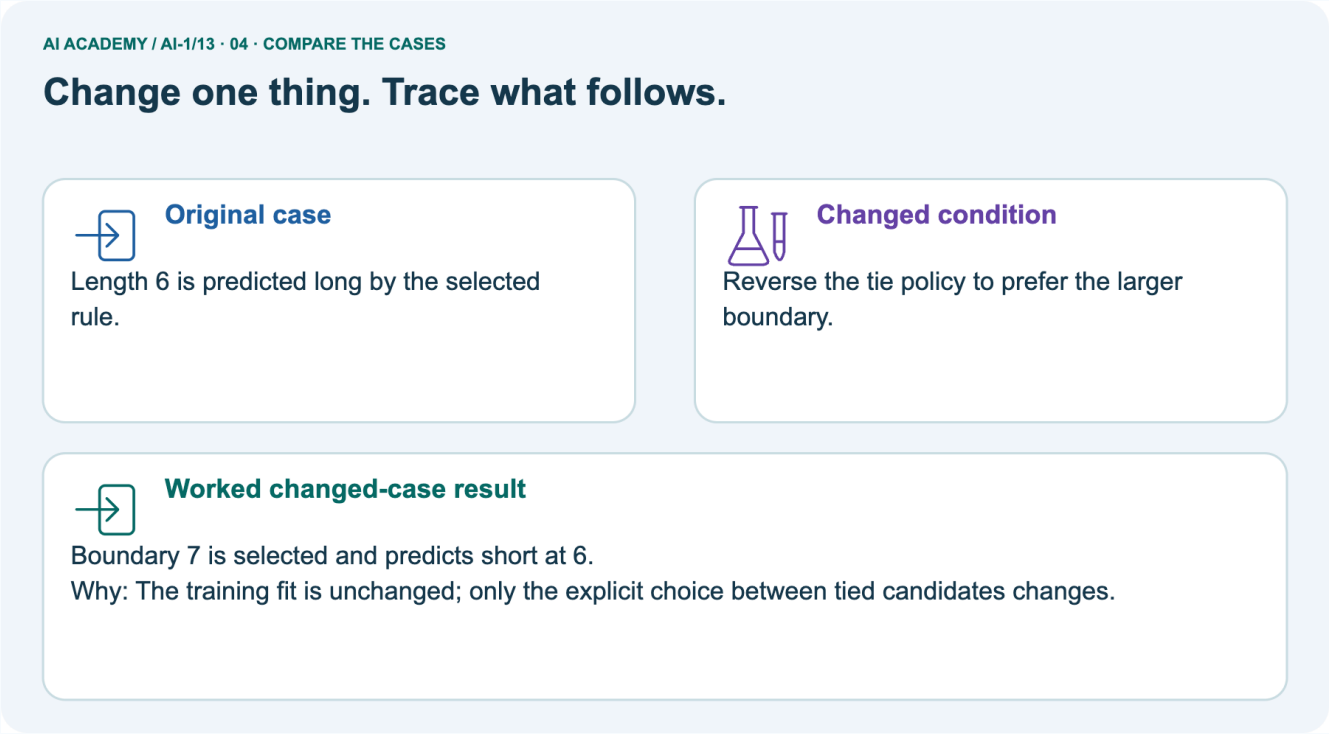
Follow the labelled picture with the matching explanation. Open the vector diagram

Same **training** errors, different new **prediction**. Candidate 5: Boundary = 5, **Training** errors = 0, **Prediction** at 6 = long. Candidate 7: Boundary = 7, **Training** errors = 0, **Prediction** at 6 = short.

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

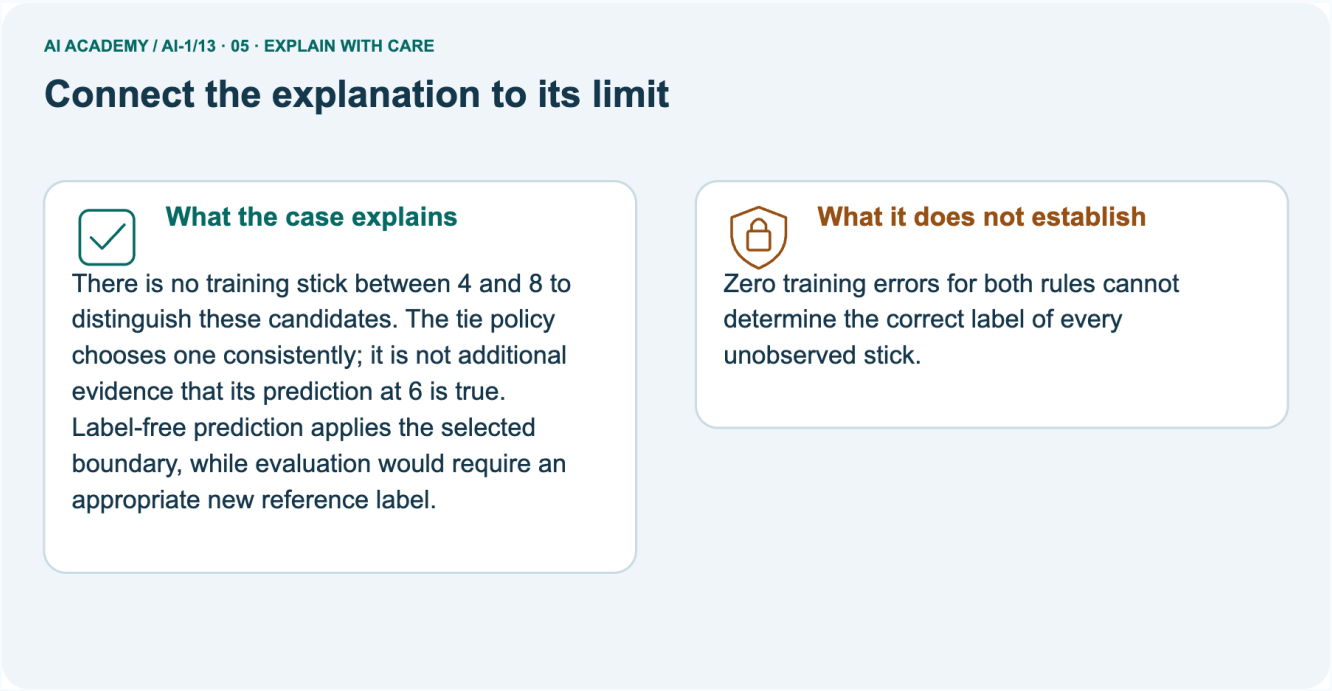
Original case: Length 6 is predicted long by the selected rule. Changed condition: Reverse the tie policy to prefer the larger boundary. Expected result: Boundary 7 is selected and predicts short at 6.

Reason: The **training** fit is unchanged; only the explicit choice between tied candidates changes.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

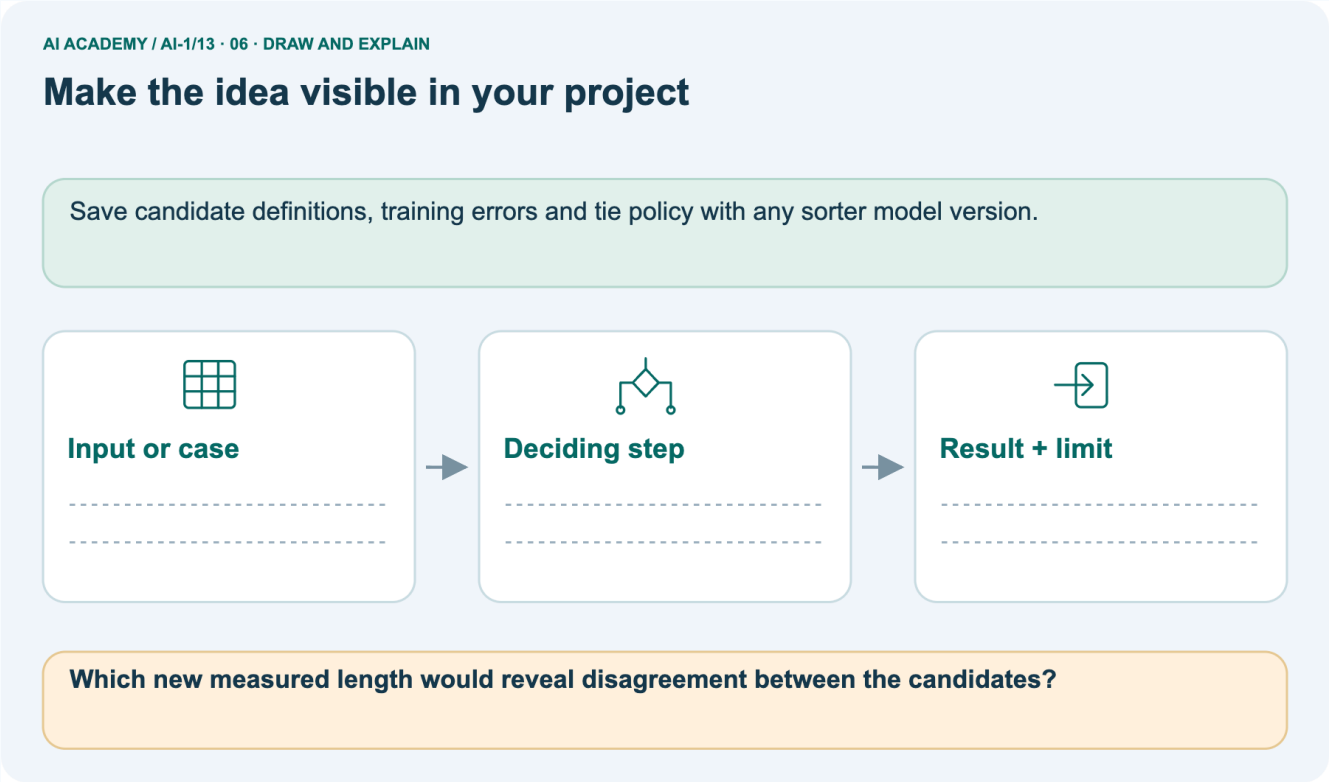
There is no **training** stick between 4 and 8 to distinguish these candidates. The tie policy chooses one consistently; it is not additional **evidence** that its **prediction** at 6 is true.

Label-free **prediction** applies the selected boundary, while evaluation would require an appropriate new reference label. Limit: Zero **training** errors for both rules cannot determine the correct label of every unobserved stick.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Save candidate definitions, **training** errors and tie policy with any sorter model version. Draw your **input**, deciding step, result and limit.

Supported question: Which new measured length would reveal disagreement between the candidates?

Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

▼ Earlier wording, alternatives and picture transcripts

01. Learning from Examples: the case

02. Follow the deciding steps

03. Same training errors, different new prediction

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.